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1  import numpy as np
2  import matplotlib.pyplot as plt
3  from scipy import stats
4  control_group = np.array([5, 7, 6, 5, 8, 7, 6, 5])
5  treatment_group = np.array([9, 10, 8, 9, 11, 10, 9, 8])
6  t_stat, p_value = stats.ttest_ind(treatment_group, control_group)
7  print("T-statistic:", t_stat)
8  print("P-value:", p_value)
9  alpha = 0.05
10 if p_value < alpha:
11     print("Result: The treatment has a statistically significant effect.")
12 else:
13     print("Result: No statistically significant difference found.")
14 labels = ["Control Group", "Treatment Group"]
15 means = [np.mean(control_group), np.mean(treatment_group)]
16 plt.bar(labels, means)
17 plt.ylabel("Average Improvement Score")
18 plt.title("Treatment vs Placebo Effect")
19 plt.text(x: 0.5, max(means), s: f"p-value = {p_value:.4f}", ha='center')
20 plt.show()
```

T-statistic: 5.779009133483481

P-value: 4.7753221801285165e-05

Result: The treatment has a statistically significant effect.

Figure 1

